SHARP FAVARD DECAY WITH UNIFORM MARGINALS

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Abstract. For the n parallelograms obtained by matching the equal subintervals of the two horizontal edges of the unit square, we prove that the minimum area of their union is of order $\frac{1}{\log n}$, as conjectured by Kaczynski.

Divide the top and bottom edges of the unit square into n equal intervals, match them by a permutation, and join matched intervals by parallelograms. Their areas add to 1; the question is how small their union can be. Writing α_n for the minimum, we call the assertion $\alpha_n \asymp \frac{1}{\log n}$ *KC2*.[1]

A strip joining I to J is encoded by the square $I \times J$ in the plane of endpoints. At height t, its section is the projection $\pi_t(I \times J)$, where $\pi_t(x, y) = (1 - t)x + ty$. A set with a unit coordinate projection has Hausdorff length at least 1. We attain this minimum with both coordinate projections equal to $[0, 1]$, together with a logarithmic bound on the averaged projections of its neighborhoods. For compact $S \subset \mathbb{R}^2$, put $S_\delta = S + [-\delta, \delta]^2$ and define its unnormalized Favard length by

$$\mathrm{Fav}(S) = \int_0^\pi |p_\theta(S)|\, d\theta, \qquad p_\theta(x, y) = x \cos\theta + y \sin\theta.$$

Let p_x, p_y be the coordinate projections. We write $|\cdot|$ for length or area, and $\mathcal{H}^1$ for Hausdorff length.

Theorem 1. *There is a compact set $S \subset [0, 1]^2$ such that*

$$p_x(S) = p_y(S) = [0, 1], \qquad \mathcal{H}^1(S) = 1,$$

and, for absolute constants $c, C > 0$,

$$\frac{c}{\log \frac{1}{\delta}} \le \mathrm{Fav}(S_\delta) \le \frac{C}{\log \frac{1}{\delta}} \qquad (0 < \delta < \frac{1}{2}).$$

Its length measure $\mu = \mathcal{H}^1|_S$ has uniform coordinate marginals.

Corollary 2 (KC2)**.** *There is an absolute constant C such that, for every $n \ge 2$,*

$$\frac{1}{H_n} \le \alpha_n \le \frac{C}{\log n}, \qquad H_n = \sum_{j=1}^{n} \frac{1}{j}.$$

The marginal condition means that $\mu(B \times [0, 1]) = \mu([0, 1] \times B) = |B|$ for Borel $B \subset [0, 1]$. A probability measure with these marginals is a *permuton* [4]. In our construction, the marginal condition follows from length 1.

Chang, Shmerkin and Suomala [1] obtain sharp Favard decay for random Cantor sets. Csörnyei and Suomala [2] characterize the measures which can occur as one projection of a purely unrectifiable length measure. We require both marginals, together with the quantitative bound. The fixed-base construction is too restrictive: for nested L-adic permutation patterns, [1, Theorem A.1] gives Favard length $\gtrsim_L \frac{\log j}{j}$ at scale L^{-j}. We instead allow the widths of the squares to vary. Throughout, c, C are positive absolute constants, allowed to change.

1. The matching reduction

Length 1 already forces the marginal condition. Indeed, suppose $p_x(S) = p_y(S) = [0, 1]$ and $\mathcal{H}^1(S) = 1$, and put $\mu = \mathcal{H}^1|_S$. For either coordinate projection p and Borel $B \subset [0, 1]$, the set $S \cap p^{-1}(B)$ projects onto B. Since p is 1-Lipschitz, $\mu(p^{-1}(B)) \ge |B|$. The same inequality for the complement, together with total mass 1, forces equality.

Put $\mathcal{A}(S) = \int_0^1 |\pi_t(S)|\, dt$. The angular change of variables $t = \frac{\sin\theta}{\cos\theta + \sin\theta}$ gives

$$\mathcal{A}(S) = \int_0^{\frac{\pi}{2}} \frac{|p_\theta(S)|}{(\cos\theta + \sin\theta)^3}\, d\theta \le \mathrm{Fav}(S). \tag{1}$$

[1] KC2 abbreviates "Kaczynski's Conjecture 2," following its numbering in Kuklinski [5, § 3].

Let μ have uniform coordinate marginals and be supported on S. Partition $[0,1]$ into n intervals I_i of length $\frac{1}{n}$, half-open except for the last, which includes 1. The matrix $a_{ij} = n\mu(I_i \times I_j)$ has every row sum and column sum equal to 1. Any k rows have total mass k and therefore meet at least k columns, since each column has mass 1. This is Hall's condition, so the matching theorem [3] supplies a permutation σ with $a_{i,\sigma(i)} > 0$ for all i. Every selected cell meets S, so, closing the cells,

$$P_\sigma := \bigcup_{i=1}^{n} \overline{I}_i \times \overline{I}_{\sigma(i)} \subset S_{\frac{1}{n}}. \tag{2}$$

The slice of the associated strip union at height t is $\pi_t(P_\sigma)$. Thus the upper bound in Theorem 1 implies the corollary through

$$\alpha_n \le \mathcal{A}(P_\sigma) \le \mathcal{A}(S_{\frac{1}{n}}) \le \mathrm{Fav}(S_{\frac{1}{n}}) \le \frac{C}{\log n}. \tag{3}$$

For $n = 2$, increase C if necessary.

The lower bound is a pairwise overlap count. In any permutation-strip arrangement of area A, let Z count the strips covering a point. Then $\int Z = 1$ and Cauchy–Schwarz gives $1 \le A \int Z^2$. Two crossing strips with endpoint index gaps $p, q \ge 1$ have intersection area

$$\frac{1}{n(p+q)} \le \frac{1}{4n}\left(\frac{1}{p} + \frac{1}{q}\right).$$

Their signed displacement is $\frac{p-(p+q)t}{n}$, so the overlap length has a triangular graph of height $\frac{1}{n}$ and base $\frac{2}{p+q}$. Noncrossing pairs have no positive-area intersection. On either edge there are $n-d$ pairs of index gap d. Counting each pair twice in Z^2 therefore gives

$$\int Z^2 \le 1 + \frac{1}{n}\sum_{d=1}^{n-1} \frac{n-d}{d} = H_n, \qquad A \ge \frac{1}{H_n}. \tag{4}$$

For any uniform-marginal measure, take $n = \lceil \frac{1}{\delta} \rceil$ in (2). Since $\frac{1}{n} \le \delta$, equations (1) and (4) give $\mathrm{Fav}(S_\delta) \ge \mathcal{A}(P_\sigma) \ge \frac{1}{H_n} \ge \frac{c}{\log \frac{1}{\delta}}$. Thus the Favard lower bound holds for every compact set carrying a probability measure with uniform marginals. We now construct a set attaining it with length 1.

2. Finite patterns

A *pattern* is a set $P = \bigcup_{i=1}^{N} (I_i \times J_i)$, where each family of closed intervals (I_i) and (J_i) covers $[0,1]$ with disjoint interiors, and $|I_i| = |J_i| = w_i > 0$. Write $N(P)$ for its number of squares. The sum of their sides is 1.

A red–blue shuffle preserves both interval partitions while reducing projection length. We also control the number of squares: thickening N squares by δ costs at most $CN\delta$ in Favard length.

Lemma 1 (Finite patterns)**.** *For every integer $k \ge 1$ there is a pattern P such that*

$$\mathrm{Fav}(P) \le \frac{C}{k}, \qquad \log N(P) \le Ck.$$

Proof. Each shuffle will give a quadratic expected saving in Favard length at bounded expected cost in $\log N$. Given a pattern P, write $E_t = \pi_t(P)$, $\ell(t) = |E_t|$ and $L(P) = \int_0^{\frac{1}{2}} \ell(t)\,dt$. We first estimate the overlap for $0 \le t \le \frac{1}{2}$.

Choose a width $0 < w < 1$ and partition a unit interval into $m = \lceil \frac{1}{w} \rceil$ pieces of width w, with the last possibly shorter. Take red and blue copies of this partition, and place them on coordinate intervals of length 2 in the orders

$$\begin{aligned} &x\text{-axis}: \quad R_1, B_1, R_2, B_2, \ldots, R_m, B_m, \\ &y\text{-axis}: \quad B_1, B_2, \ldots, B_m, R_1, R_2, \ldots, R_m. \end{aligned} \tag{5}$$

Equal labels determine squares. Put a homothetic copy of P inside each square, then divide both coordinates by 2. The resulting pattern P_w has $MN(P)$ squares, where $M = 2m$. Its copy ratios sum to 1, so subadditivity gives $|\pi(P_w)| \le |\pi(P)|$ for every linear projection π.

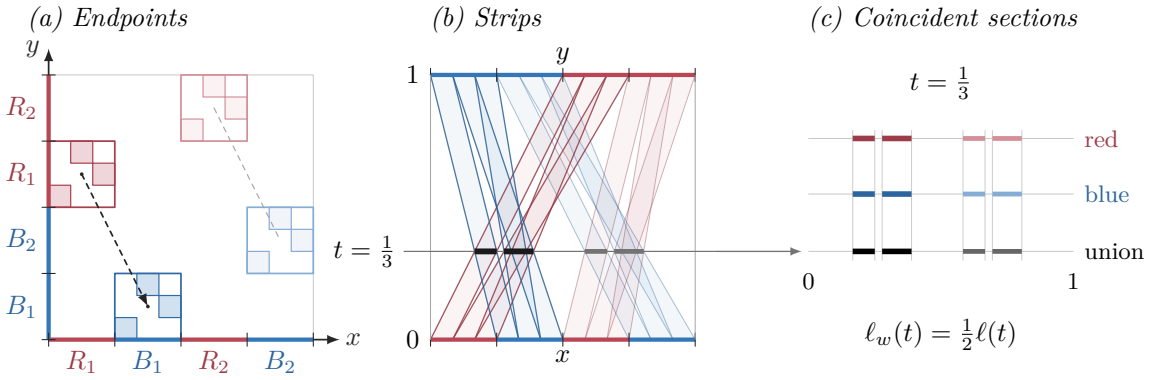


FIGURE 1. The shuffle for $w = \frac{1}{2}$. Each labeled square contains the same three-square pattern; the first red–blue pair is emphasized. Its translation follows a fiber of $\pi_{\frac{1}{3}}$, so the paired copies have identical sections at $t = \frac{1}{3}$. In (c), the red and blue sections are drawn separately; their union is the section in (b).

Within either color, the enclosing squares occur in the same order on both axes, so their π_t-projections have disjoint interiors. Each color contributes exactly $\frac{\ell(t)}{2}$. Thus, writing $\ell_w(t) = |\pi_t(P_w)|$, we have $\ell_w(t) \le \ell(t)$, and the difference is the overlap between the colors.

For a full-width pair, blue starts w to the right of red on the x-axis and one unit to its left on the y-axis, before scaling. The projected displacement, in units of their common side, is $d_w(t) = (1-t) - \frac{t}{w}$. The full-width pieces occupy at least half of each unit partition. Their red squares therefore have total side length at least $\frac{1}{4}$ after scaling. Adding the overlaps with their blue partners gives

$$\ell(t) - \ell_w(t) \ge \frac{1}{4} \left| E_t \cap (E_t + d_w(t)) \right|. \tag{6}$$

There is no overcounting because the red projections have disjoint interiors.

Averaging over translations makes the overlap depend only on length: for measurable $E \subset [0,1]$, Fubini gives [6]

$$\int_{\mathbb{R}} |E \cap (E+s)|\, ds = |E|^2. \tag{7}$$

The integral counts the pairs in $E \times E$. Its integrand is even and vanishes outside $[-1,1]$.

For $0 < t \le \frac{1}{2}$, a negative shift $d_w(t) = -s$, $0 \le s \le 1$, requires $w = \frac{t}{1-t+s} \asymp t$. Small heights need small widths, so we sample $w = U^2$, with U uniform on $(0,1)$. Then

$$U = \sqrt{\frac{t}{1-t+s}}, \qquad |dU| = \frac{\sqrt{t}}{2(1-t+s)^{\frac{3}{2}}}\, ds \gtrsim \sqrt{t}\, ds.$$

These values lie in $(0,1]$. Evenness and (7) give $\mathbb{E}_w[\ell(t) - \ell_w(t)] \gtrsim \sqrt{t}\, \ell(t)^2$. The square law gives an integrable reciprocal weight: $\int_0^{\frac{1}{2}} \frac{dt}{\sqrt{t}} < \infty$. Weighted Cauchy–Schwarz gives

$$\mathbb{E}_w[\mathcal{A}(P) - \mathcal{A}(P_w)] \gtrsim \int_0^{\frac{1}{2}} \sqrt{t}\, \ell(t)^2\, dt \gtrsim \left(\int_0^{\frac{1}{2}} \ell(t)\, dt \right)^2 = L(P)^2. \tag{8}$$

Since the weight in (1) is at most 1, the saving in Favard length is at least the saving in $\mathcal{A}$.

To cover all directions, let $T(x,y) = (y,x)$ and $R(x,y) = (x, 1-y)$, and first choose G uniformly from $\{I, T, R, T \circ R\}$. The map T exchanges t and $1-t$, while R supplies the other angular quadrant. Hence

$$\sum_G L(GP) = \mathcal{A}(P) + \mathcal{A}(RP) \asymp \mathrm{Fav}(P). \tag{9}$$

Apply the shuffle to GP. Favard length is invariant under G; averaging (8) and using Cauchy–Schwarz in (9) gives the quadratic saving

$$\mathbb{E}[\mathrm{Fav}(P) - \mathrm{Fav}((GP)_w)] \ge c\, \mathrm{Fav}(P)^2.$$

Each shuffle multiplies the number of squares by M, so its contribution to $\log N$ is $\log M$. Since $M = 2\lceil \frac{1}{w} \rceil \le \frac{4}{w} = \frac{4}{U^2}$, $\mathbb{E}\log M \le \log 4 + 2\int_0^1 \log \frac{1}{u}\, du < \infty$.

Starting with the unit square, use fresh independent choices of G, U at each of k steps. Let F_k, N_k be the resulting Favard length and number of squares, and put $f_k = \mathbb{E}F_k$. Conditional expectation and Jensen's inequality yield

$$f_{k+1} \le f_k - cf_k^2, \qquad f_k \le \frac{4}{1+4ck}, \qquad \mathbb{E}\log N_k \le Ck.$$

Here $f_0 = 4$, and $\frac{1}{f_{k+1}} \ge \frac{1}{f_k} + c$ proves the middle bound. To obtain both bounds for the same pattern, use their normalized sum: $\mathbb{E}[kF_k + \frac{\log N_k}{k}] \le C$. Some realization has this sum at most C; since both terms are nonnegative, it satisfies both estimates in the lemma. □

In particular, choosing k of order 2^j supplies patterns P_j with

$$\mathrm{Fav}(P_j) \le 2^{-j}, \qquad \log N(P_j) \le C2^j. \tag{10}$$

3. The limiting set

To nest patterns, insert a copy of Q into each square of $P = \bigcup_i (v_i + w_i[0,1]^2)$:

$$P \star Q = \bigcup_i (v_i + w_i Q).$$

The result is again a pattern, with $N(P)N(Q)$ squares. Moreover, $P \star Q \subset P$, while $\sum_i w_i = 1$ gives, for every linear projection π,

$$|\pi(P \star Q)| \le \min\{|\pi(P)|, |\pi(Q)|\}. \tag{11}$$

The squares of a pattern have total diameter $\sqrt{2}$. To obtain length 1, we group smaller squares inside copies of a pattern whose diameter is close to 1. For even m, let B_m be the equal-grid pattern given by the cyclic half-shift $i \mapsto i + \frac{m}{2} \pmod m$. It lies in the diamond

$$|x - \frac{1}{2}| + |y - \frac{1}{2}| \le \frac{1}{2} + \frac{1}{m}.$$

The diamond has diameter $1 + \frac{2}{m}$, so

$$\operatorname{diam} B_m \le 1 + \frac{2}{m}. \tag{12}$$

Thus insertion into B_m preserves the projection bound and gives a cover of diameter tending to 1.

(a) The half-shift *(b) Insertion into a pattern*

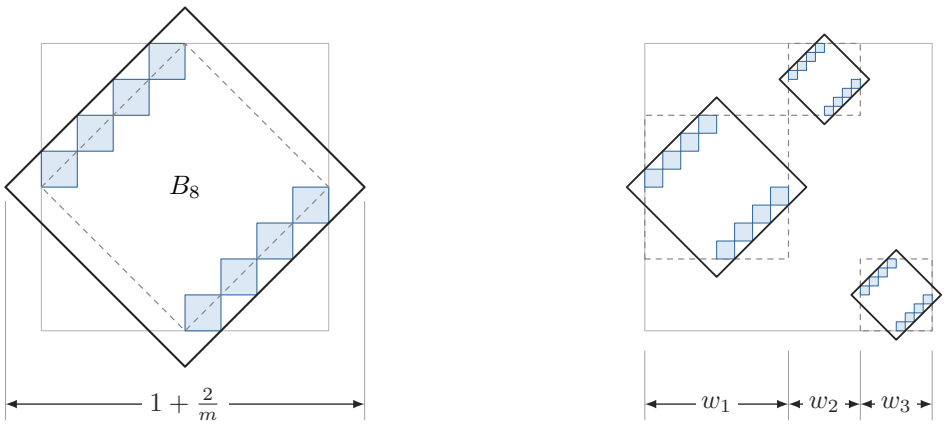


FIGURE 2. The half-shift B_8 and its insertion into three squares of widths $w_1 = \frac{1}{2}$, $w_2 = w_3 = \frac{1}{4}$. The dashed diamond has diameter 1. The solid diamonds enclose the copies; on the right their total diameter is $\left(1 + \frac{2}{m}\right)\sum_i w_i = 1 + \frac{2}{m}$, with $m = 8$.

Set $Q_j = B_{2j} \star P_j$. Form the nested patterns $T_0 = [0,1]^2$ and $T_j = T_{j-1} \star Q_j$. Equations (10)–(12) give

$$\mathrm{Fav}(T_j) \le C2^{-j}, \qquad \log N(T_j) \le C2^j. \tag{13}$$

For the second estimate, sum the logarithmic costs of the factors: $\sum_{i \le j} (\log(2i) + C2^i) \le C2^j$. The factor B_{2j} makes every child square at most half as wide as its parent, so the largest square side in T_j is at most 2^{-j}.

Let $S = \bigcap_j T_j$. The sets $v_i + w_i B_{2(j+1)}$, indexed by the squares $v_i + w_i[0,1]^2$ of T_j, cover S. Their diameters tend uniformly to zero, and their total diameter is at most $\sum_i w_i(1 + \frac{1}{j+1}) = 1 + \frac{1}{j+1}$. Thus $\mathcal{H}^1(S) \le 1$. Both coordinate projections of S are $[0,1]$, by nested compactness, so $\mathcal{H}^1(S) \ge 1$ as well.

It remains to check the rate between successive construction scales. Enlarging a union of N squares by $[-\delta, \delta]^2$ increases its Favard length by at most $8N\delta$: in direction θ, each projected interval gains $2\delta(|\cos\theta| + |\sin\theta|)$. For a fixed absolute $B > 0$, equation (13) gives

$$\mathrm{Fav}(S_\delta) \le C2^{-j} + 8e^{B2^j}\delta.$$

For small δ, choose the largest j with $e^{B2^j} \le \delta^{-\frac{1}{2}}$. This keeps the thickening error at most $8\sqrt{\delta}$, while $2^j \asymp \log \frac{1}{\delta}$. Thus

$$\mathrm{Fav}(S_\delta) \le \frac{C}{\log \frac{1}{\delta}} + 8\sqrt{\delta} \le \frac{C}{\log \frac{1}{\delta}}.$$

The remaining values of δ are covered by increasing C. Section 1 supplies the uniform marginals and the matching lower estimate, completing the proof. In particular, $\mathrm{Fav}(S) = 0$ and $\mathcal{H}^1(S) = 1$; the Besicovitch projection theorem implies that S is purely unrectifiable.

Conversely, KC2 supplies equal-grid patterns with $\mathcal{A}(P_j) \le 2^{-j}$ and $\log N(P_j) \le C2^j$: take $n_j = \lceil \exp(C_* 2^j) \rceil$. Equations (1) and (11) give $\mathrm{Fav}(P_j \star R(P_j)) \le C\mathcal{A}(P_j)$: the two factors control the two angular quadrants. Thus KC2 also gives (10), after shifting j by a fixed amount. The same half-shifts and nested insertions give Theorem 1. The theorem and KC2 are therefore equivalent.

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